Resolution Geometry: Why Monotonicity, Why Fisher, and Where the Lorentz Sign Comes From

Conditional Naturality from Simulation and Descent, Established Frameworks for the Sign, and a Corrected Ledger of Three Different Signs

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Companion to *Monotonicity, Hessian and Lorentz-Sign Protocol* (release package)

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Abstract

Statistics, thermodynamics, quantum information and estimation theory all end up with monotone information measures and, locally, with the Fisher metric. We explain why, and we locate exactly where the Lorentz sign can and cannot come from.

Two roots, not one axiom.

- *Simulation*: whatever can be decided from a garbled record can be decided from the original. This forces monotonicity of every value-of-information functional (Blackwell), and hence of every f-divergence, as a mixture of decision values.
- *Descent*: two-sided invariance under lossless re-descriptions forces Fisher, by Čencov's theorem derived from RG descent. Lossless re-descriptions are exactly the congruent splittings.

Descent is strictly weaker than monotonicity under all channels, so Čencov's selection is deeper than the monotonicity condition itself. Local asymptotic normality (LAN) then explains why *measurements* incline to Fisher: every regular experiment becomes, at scale $n^{-1/2}$, a Gaussian shift with Fisher covariance.

The sign. Fisher is positive semidefinite by construction. No pullback, restriction, sum or nuisance reduction makes it indefinite, and a negative "observed information" is not Lorentzian. A Lorentz sign arises from three established structures:

(i) hyperbolic polynomials and Koszul–Vinberg characteristic functions: on the Lorentz cone the Fisher metric of the cone exponential family is $\operatorname{Hess}\log Z$, and its radial reflection is $-(\alpha/q)\operatorname{Hess} q$;

(ii) the qubit state cone, which *is* the $1+3$ Lorentz cone ($\det(tI + x\cdot\sigma) = t^2 - |x|^2$, preserved by $SL(2,\mathbb{C})$);

(iii) hyperbolic propagation, where Lorentz inertia is one declared negative direction plus a positive transverse Schur complement.

Monotone geometry supplies exactly the $d-1$ positive directions. The single negative direction needs one non-monotone declaration: a clock, a radial direction or a propagation direction.

The slipped sign. The supplied battery issued one wrong-sign curvature certificate. We show that it is a Type-S error at near-null truth ($|R|/\mathrm{se} = 0.47$). The expected number over the admitted designs is 1.44; 1 was observed, and $P(\geq 1) = 0.76$. It is therefore statistical, not structural. A ledger separating the Fisher sign, the signature convention and the curvature sign makes such slips impossible to misread.

Evidence. The battery is deterministic and exact where possible, with no random sampling.

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1 The puzzle

Why do unrelated frameworks agree? Consider:

- the second law, as relative entropy to equilibrium decreasing along any dynamics that fixes equilibrium;
- the quantum data-processing inequality under CPTP maps;
- the information inequality behind Cramér–Rao bounds;
- the monotonicity of every f-divergence.

Locally, every one of them, through its Hessian, produces Fisher. Agreement across frameworks with different axioms suggests a common root beneath monotonicity itself. We identify two roots and state precisely which conclusions need which.

2 Root 1: simulation forces monotonicity

Definition 2.1 (Value of information)**.** For a decision problem $(\Theta, \mathcal{A}, L, \pi)$ and an experiment $P(\cdot \mid \theta)$, the Bayes risk is $r(P) = \min_\delta \sum_\theta \pi(\theta) \sum_y P(y \mid \theta) L(\theta, \delta(y))$. The value of information is $V(P) = r(\text{no data}) - r(P)$.

Theorem 2.2 (Simulation; Blackwell [1])**.** *For every channel K and every decision problem, $r(KP) \geq r(P)$, equivalently $V(KP) \leq V(P)$. Conversely, if $V(Q) \leq V(P)$ for every decision problem, then Q is a garbling of P (Blackwell–Sherman–Stein).*

Proof of the forward direction. Any rule δ' applied to the output of K is a randomized rule on the original record: first draw from $K(\cdot \mid y)$, then apply δ'. The minimum over a larger set of rules cannot be larger. □

Witness (R1). Nine exact rational decision problems (three priors, three loss tables) under a fixed $3 \to 2$ garbling. Bayes risk never decreases.

Imported Theorem 2.3 (f-divergences are mixtures of decision values [2])**.** *Every f-divergence is an integral, over prior weights, of statistical informations (binary Bayes-risk gaps), with a nonnegative weight determined by f''.*

Corollary 2.4 (Why monotonicity is everywhere)**.** *Every f-divergence is monotone under every channel because it is a nonnegative mixture of decision values, each of which is monotone by simulation. Monotonicity is not an axiom of each framework. It is inherited from "less data can be simulated by more data".*

3 Root 2: descent forces Fisher, and is deeper than monotonicity

Imported Theorem 3.1 (Čencov from descent, from the companion print [13])**.** *A Markov map has a Markov left inverse iff it is a congruent splitting. A continuous positive metric family that is invariant under these lossless re-descriptions is c·Fisher, with one constant in every dimension.*

Proposition 3.2 (The hierarchy of hypotheses)**.**

$$\begin{aligned} \textit{simulation order} &\implies \textit{monotone under all channels} \\ &\implies \textit{two-sided invariance under lossless re-descriptions (descent).} \end{aligned}$$

Descent is the formally weakest hypothesis: it is an equality *on a smaller class of maps, rather than an* inequality *on all maps. Fisher selection needs only descent, together with continuity and positivity.*

Proof. The first implication is Theorem 2.2. For the second, a channel K with a Markov left inverse M gives $h_p(u) \geq h_{Kp}(Ku) \geq h_{MKp}(MKu) = h_p(u)$.

For continuous positive metrics, the Čencov print shows that descent already gives c·Fisher, which is monotone, so the two hypotheses become equivalent *as a consequence*. Whether descent without continuity implies monotonicity is the open continuity question of the Čencov print.

Within-dimension descent is not enough. The family $c(n)F$ with c increasing in n is permutation-invariant, but it violates splittings across dimensions, and it is not monotone under them. □

Reading. Čencov's selection rests on descent, an invariance, rather than on monotonicity, an inequality. Monotonicity of Fisher is then a theorem: the defect is a conditional score variance.

4 Where the frameworks meet

4.1 Estimation: why measurements incline to Fisher

Imported Theorem 4.1 (Local asymptotic normality [3, 4])**.** *For a regular i.i.d. model, the log-likelihood ratio at $\theta + h/\sqrt{n}$ converges to $\mathcal{N}(-\frac{1}{2}h^\top Fh,\ h^\top Fh)$. Every regular experiment becomes locally a Gaussian shift with Fisher covariance.*

Witness (R3), Bernoulli $\theta = 0.3$, $h = 1$, exact binomial sums. The mean and variance of the log-likelihood ratio are $(-2.160, 4.100)$, $(-2.295, 4.504)$ and $(-2.336, 4.626)$ at $n = 100$, 1000 and 4000, against $(-2.381, 4.762)$. The error halves as n quadruples, i.e. it shrinks like $n^{-1/2}$. *This is why practice inclines to Fisher.* Whatever the model, enough repetitions make Fisher the only surviving local structure, and every admissible divergence has Hessian c·Fisher (Hessian print).

4.2 Thermodynamics: the second law as data processing

Proposition 4.2 (H-theorem)**.** *If a Markov dynamics T fixes π, then $D(Tp\|\pi) = D(Tp\|T\pi) \leq D(p\|\pi)$. Free energy relative to equilibrium is $kT\,D(p\|\pi)$ plus a constant, and it never increases.*

Witness (R2). An exact rational chain fixing $\pi = (\frac{1}{2}, \frac{1}{3}, \frac{1}{6})$ gives $D = 0.974$, 0.268, 0.0749, 0.0203, and so on.
The Gibbs Fisher information is $\mathrm{Var}(E) = k_B T^2 C_V$ (Hessian print). *Limit:* time is not a data-processing order. Invertible dynamics forgets nothing (Rooting Closure), and the monotone direction exists only for dynamics that fix a reference law.

4.3 Quantum: monotone, but not unique

CPTP monotonicity of relative entropy [5, 6] is Theorem 2.2 with complete positivity, RG's extension gate. Descent keeps every Petz metric and selects none, so the task declares the member (Quantum Mechanics print).

4.4 Conditional naturality

Theorem 4.3 (Conditional naturality of monotone geometry)**.** *Consider a finite or dominated classical experiment, with regular smooth models and a positive, continuous measure of distinguishability that respects simulation, or merely descent.*

(a) It is monotone under garbling (Theorem 2.2).

(b) Its local form is c·Fisher (Čencov from descent, or the Hessian theorem for divergences).

(c) In the large-sample limit, every regular experiment is locally a Gaussian shift with Fisher covariance (LAN).

In quantum theory, (a) holds under CPTP maps and (b) is replaced by the Petz band.

5 Fisher cannot be Lorentzian

No-Go 5.1 (No sign from Fisher operations). Pullbacks, restrictions, positive sums and nuisance minimizations of positive Fisher tensors are positive semidefinite. Real invertible reparametrizations preserve inertia. The isometry group of a positive form is compact, whereas boosts are not, so no positive Fisher form is boost-invariant on the same tangent representation.

No-Go 5.2 (Observed information is not a signature). Observed information, minus the Hessian of the log-likelihood, can be negative away from the maximum, while Fisher information is not. For Cauchy location with data ± 3, the observed information at $\theta = 0$ is $-\frac{8}{25}$, while Fisher is $\frac{1}{2}$ per point (R7). A negative eigenvalue from a numerical Hessian is a diagnostic of evaluation away from the optimum, or of numerical error. It is never evidence of Lorentz signature.

6 Where the sign comes from: three established structures

6.1 Hyperbolic polynomials and Koszul–Vinberg cones

Imported Theorem 6.1 (Gårding; Koszul–Vinberg; barriers [7, 8, 9, 10])**.** *For a hyperbolic polynomial q of degree m with hyperbolicity cone C, the function $q^{1/m}$ is concave on C, so $-\log q$ is a convex, logarithmically homogeneous barrier. For a regular convex cone, the characteristic function $Z(z) = \int_{C^*} e^{-\langle z,s\rangle} ds$ has $\operatorname{Hess} \log Z \succ 0$, invariant under $\operatorname{Aut}(C)$. For the Lorentz cone, $Z \propto q^{-d/2}$.*

Theorem 6.2 (Radial reflection; companion protocol, Theorem 4.2)**.** *Let $\Psi = -\alpha \log q$, with $q > 0$ two-homogeneous on a cone and $F = \operatorname{Hess} \Psi \succ 0$. Then*

$$g = F - \alpha^{-1} d\Psi \otimes d\Psi = -\frac{\alpha}{q} \operatorname{Hess} q$$

has inertia $(1, d-1)$. The negative direction is the radial one, and its F-orthogonal complement is positive.

Witness (R4), exact. For the $1+1$ Lorentz cone, $Z = 2/(t^2 - x^2)$. The Fisher metric $\operatorname{Hess} \log Z$ is positive definite, and the radial reflection equals $-(d/q)J$ with $J = \operatorname{diag}(1, -1)$, symbolically. *Reading.* Fisher stays positive. A *new* form is built by subtracting the declared radial direction. Monotone geometry supplies the positive complement, and the declaration supplies the minus sign.

6.2 The qubit cone is the Lorentz cone

Proposition 6.3 (R5)**.** *For $\rho \propto tI + x \cdot \sigma$:*

- $\det = t^2 - |x|^2$*, so positivity of the state is the forward-cone condition;*
- $SL(2, \mathbb{C})$ *acting by $\rho \mapsto A\rho A^\dagger$ preserves* det*, which is the spinor double cover of the Lorentz group;*
- $\operatorname{Hess}(-\log \det) \succ 0$*, the canonical barrier, i.e. monotone-type geometry;*
- $\operatorname{Hess} \det = 2 \operatorname{diag}(1, -1, -1, -1)$.

All four are verified symbolically.

Reading. The positive cone of the smallest quantum system and the Minkowski cone are one Jordan-algebraic object, the spin factor [10, 11]. This is a kinematic identity of cones. It does not derive spacetime, but it is an established place where quantum positivity and Lorentz signature meet.

6.3 Hyperbolic propagation

Theorem 6.4 (Lorentz inertia as one negative plus a positive block; R6)**.** *Write*

$$Q(\omega, k) = a\omega^2 + 2\omega\, b \cdot k + k^\top C k, \qquad a < 0.$$

Then $Q = a(\omega + b \cdot k/a)^2 + k^\top (C - bb^\top/a)k$*. So* Q *is hyperbolic in* ω*, with Lorentz inertia, iff the transverse Schur complement* $C - bb^\top/a$ *is positive definite.*

Where monotonicity enters. The transverse block is a Schur complement. When it is identified with an efficient-information or energy form, positivity is exactly what monotone geometry guarantees. The one negative coefficient a is the propagation or clock declaration.

6.4 Partial Legendre saddles

A partial Legendre transform of a convex potential in r variables has exactly r negative directions (R9: eigenvalues of opposite signs in one variable). Thermodynamic potentials such as enthalpy or Gibbs functions therefore carry signed Hessians. They are saddles, not causal cones, unless a propagation law identifies them.

6.5 The sign ledger

Theorem 6.5 (Exactly one non-monotone ingredient)**.** *In all three structures above, Lorentz inertia* $(1, d-1)$ *equals one declared negative direction plus a positive transverse form. The declared direction is radial/clock in the Koszul–Vinberg construction, the trace direction for the qubit cone, and the propagation direction for hyperbolicity. Monotone (Fisher-type) geometry supplies the positive block and nothing else. Signature does not fix the curvature sign: within one Lorentzian family, positive, zero and negative curvature all occur with positive Fisher information (companion protocol, Corollary 7.5).*

Proof. Theorems 6.2 and 6.4, R5, and inertia additivity. □

7 The slipped sign, and three signs that must not be conflated

Sign	What determines it	Can it be negative?	Typical slip
Fisher eigenvalue	score covariance	never; a negative value is a bug	observed information away from the maximum (No-Go 5.2); finite-difference error
Signature convention	choice of mostly-plus or mostly-minus	overall sign is a convention	$g = -(\alpha/q)\,\mathrm{Hess}\, q$ has the opposite overall sign to $\mathrm{Hess}\, q = 2J$; both are Lorentzian
Curvature sign	physics and data	yes, as a measurement	Type-S decision error at near-null truth

Convention check. With $R^a{}_{bcd} = \partial_c \Gamma^a_{db} - \dots$ and $ds^2 = -N^2 dt^2 + dx^2$, a symbolic computation gives $R^x{}_{txt} = NN''$ and $R = -2N''/N$, as used in the companion. A positive tidal coefficient $E = N''/N$ therefore means *negative* scalar curvature in $1+1$ under this convention, while the tidal acceleration is $-E\xi$, as in the Newtonian limit $-\partial^2\Phi\,\xi$. This is a convention, not a slip, and it must be stated whenever a curvature sign is reported.

What actually slipped (R8). The supplied identification battery issued one wrong-sign curvature certificate among 1,668. It occurred in the design with $L = 0.3$, $b = 0$ and exposure 10^5, where the true $R = -2a^2 = -0.045$ and se $= 0.096$, so $|R|/\text{se} = 0.47$. A 95% interval excludes zero on the wrong side with probability

$$P_S = \Phi(-z_{0.975} - |R|/\text{se}) = 0.0076$$

per trial. Summed over all admitted designs, the expected number of wrong-sign certificates is 1.44. One was observed, and $P(\geq 1) = 0.76$. The slip is a statistical Type-S error [12], exactly where the truth is close to zero. It is not an error in the derivation, the signature or the convention.

The mature rule.

1. Issue a sign certificate only when a truth-independent gate bounds $|R|/\text{se}$ from below over the declared parameter box. The companion's uniform gate does this: 0 wrong of 1,382 issued, which is not a zero-risk guarantee.
2. Report $P_S \leq \Phi(-z - \underline{\text{snr}})$ with each certificate.
3. Abstain otherwise.
4. Never read a negative numerical Hessian eigenvalue as a signature.

8 Typed ledger

Statement	Status	Root
Monotonicity of value-of-information and f-divergences	FORCED	simulation (Blackwell; Liese–Vajda)
Fisher as the local geometry (classical)	FORCED	descent (Čencov print), Hessian print
Fisher as the large-sample universal	FORCED (regular models)	LAN
Second law as monotone free energy	FORCED given $T\pi = \pi$	data processing
Quantum monotonicity	FORCED	simulation with complete positivity
Quantum local metric	DECLARED (Petz band)	task lock
Lorentz signature	CONDITIONAL	one declared non-monotone direction plus a positive block
Curvature sign	MEASURED	data, with a Type-S risk

9 Battery, companion results, and limits

This paper. `rg_monotone_root.py` runs R1–R9, which are exact, symbolic or deterministic. Reruns are byte-identical.

- R1: Bayes risk under garbling.
- R2: an H-theorem chain.

- R3: LAN from exact binomial sums.
- R4: the Lorentz-cone characteristic function and its radial reflection.
- R5: the qubit cone.
- R6: completion of the square.
- R7: Cauchy observed information.
- R8: the Type-S analysis of the supplied results.
- R9: a partial Legendre saddle.

Companion protocol package, rerun here. All five batteries and the verifier pass: 50/50 assertion groups and 24/24 consistency checks. The quadratic-form classification covers 250 forms (100 elliptic, 90 Lorentzian, 60 ultrahyperbolic) with no mismatches. There are 100 cone reconstructions and 3,600 synthetic clock fits. The oracle gate gives curvature coverage 0.948 with one wrong-sign certificate. The uniform gate gives coverage 0.947 with zero wrong-sign certificates.

Limits.

- Blackwell, Liese–Vajda, LAN, Gårding, Koszul–Vinberg, the barrier theory and the CPTP monotonicity theorems are imported.
- The qubit-cone identity is kinematic, not a derivation of spacetime.
- The Lorentz sign is conditional on a declared direction.
- All statistical examples are synthetic.
- No empirical claim about spacetime is made.

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